

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 June 2026
Team ID	SWTID-2026-5356
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import India's Agricultural Crop Production dataset (1997–2021) containing State, District, Crop, Season, Area, Production and Year.
FR-2	Data Processing	Clean, validate, and organize datasets for analysis.
FR-3	SQL Analysis	Create MySQL database and execute SQL queries to analyze crop production, area cultivated, seasonal trends, and state-wise performance.
FR-4	Dashboard Visualization	Develop an interactive Tableau dashboard with KPI cards, bar charts, line charts, maps, filters, and production analysis.
FR-5	Story Visualization	Develop Tableau Story to present insights in a sequential format.
FR-6	Dashboard Integration	Embed Tableau Dashboard and Story into the Flask web application.
FR-7	User Interaction	Allow users to explore dashboards using filters, maps, and interactive visualizations.
FR-8	Project Repository	Store source code, datasets, SQL scripts, documentation, and project files in GitHub.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a user-friendly interface with interactive dashboards, filters, and easy navigation for analyzing agricultural crop production data.
NFR-2	Performance	Ensure fast loading of dashboards and smooth visualization of large agricultural datasets (1997–2021).
NFR-3	Reliability	Provide accurate data processing, consistent dashboard performance, and reliable agricultural insights
NFR-4	Availability	Dashboard and project accessible through Tableau Public and GitHub.
NFR-5	Scalability	Architecture supports adding new datasets, dashboards, and analytical features.
NFR-6	Maintainability	Modular project structure for easy updates and future enhancements.
NFR-7	Security	Secure project hosting using GitHub and HTTPS-enabled Tableau Public links.